



Drilling Hydraulics Design

Engineering Wellbore Pressures, Fluid Flow and Bit Hydraulics
for Safe, Efficient



Drilling for Non-Drilling Personnel

This intensive 3-day Drilling Hydraulics Design course provides a practical, calculation-driven understanding of how pressures behave in the wellbore under both static and dynamic drilling conditions. Building from core fluid and gas properties, participants work through hydrostatic pressure, buoyancy and axial stresses in submerged tubulars, before moving into fluid-flow fundamentals and rheological models used in modern drilling operations. The course then applies these concepts to real-world challenges such as frictional pressure losses, hydraulic horsepower, bit nozzle optimization, surge and swab effects, and hole cleaning in vertical, directional and horizontal wells. Throughout the program, worked examples and field-oriented exercises help attendees link equations and models directly to operational decisions, enabling them to design safer hydraulic programs, reduce non-productive time and support more efficient drilling performance.



LOCATION

London

Houston

Dubai

LONDON PETRO ACADEMY
Empowering Professionals for a Sustainable Future



COURSE OBJECTIVES

By the end of this course, participants will be able to calculate hydrostatic and annular pressures for complex fluid columns and understand how changes in mud weight, density and compressibility affect wellbore stability. They will be able to determine buoyancy factors and axial stresses in drillstring and casing, construct free-body diagrams, and apply basic laws of fluid flow to real drilling systems. Participants will understand and use common rheological models to estimate frictional pressure losses, hydraulic horsepower at the bit and optimum bit nozzle sizes across a range of operating conditions. They will be able to estimate pore pressure and fracture gradients, interpret leak-off test data, and relate these to safe operating windows and well control margins. In addition, attendees will be equipped to evaluate hole cleaning performance in vertical and deviated wells, assess the impact of flow rate and cuttings transport, and determine safe tripping speeds by analysing surge and swab pressures.



This course is designed for drilling personnel who are directly involved in planning, executing or supervising drilling operations and who need a deeper, engineering-level understanding of drilling hydraulics. It is particularly suitable for drilling crews, rig managers, drilling foremen, drilling and mud engineers, wellsite geologists, directional drillers and drilling contractor staff seeking to strengthen their ability to interpret and act on hydraulics data. The course is also valuable for engineers and technical professionals in operating and service companies who support well planning, real-time monitoring, and well control, and who must translate hydraulic calculations into safe, cost-effective drilling parameters. It is appropriate for both early-career engineers looking to consolidate their fundamentals and experienced staff who want to refresh and update their hydraulics skills in line with current field practices.

Day 1

Introduction to Oil & Gas Drilling

- Oil and gas development
- Formations, rock pressure, fluid traps
- Fundamentals of petroleum
- Basic geology
- Drilling hydraulics

Drilling fluid properties

- Density
- Compressibility of fluids
- Properties of gases
- Z-Factor
- Hydrostatic pressure
- Annular pressure in complex fluid columns

Fluid Mechanics

- Free body diagram
- Buoyancy
- Forces and stresses in submerged tubular

Well Control

- Identification of a kick
- Shut-in pressures
- Kill mud weight calculations
- Annular pressure during well control
- Strength of casing shoe and maximum allowable annular pressure
- Exercise



Day 2

Formation Pressure

- Pre-pressure development
- Porosity and permeability
- Pore pressure measurements
- Fluid pressure gradient and mud weight requirements
- Fluid level in annulus after loss circulation

Exercises

Prediction of Fracture Gradient

- Definition
- Estimate fracture gradient from correlations
- Measurement of fracture gradient – Leak of Test (LOT)
- LOT data Analysis
- Exercises

Basic Laws of Fluid Flow

- Introduction
- Conservation of mass
- Velocity equation
- Energy and pressure balance equations
- Hydraulic calculations
- Pressure drop through bit nozzles

Rheological Models

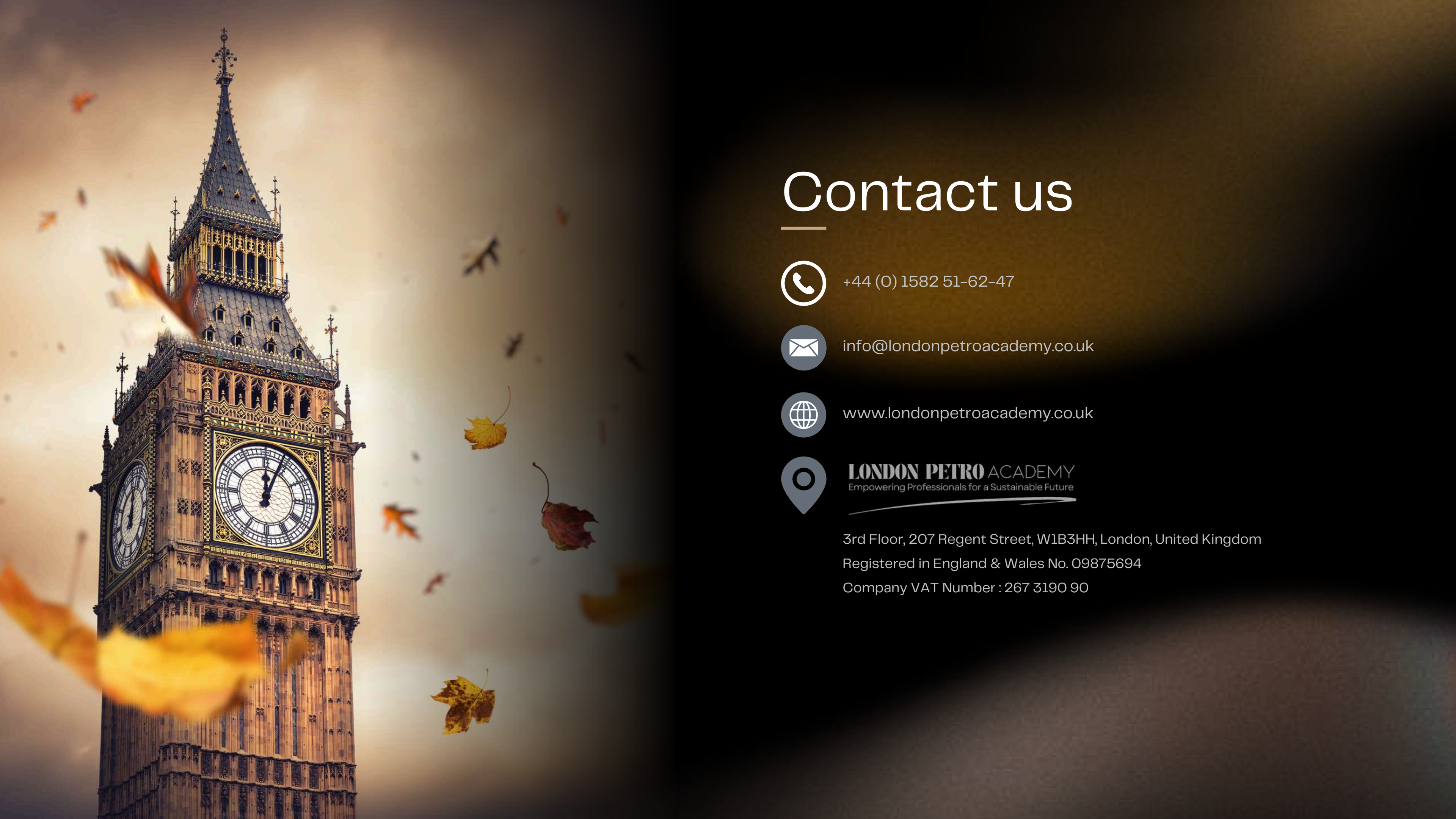
- Introduction
- Calculation of frictional pressure losses
- Newtonian model
- Bingham Plastic model
- Power-Law model
- Flow regimes, laminar and turbulent
- Frictional pressure loss equations
- Exercises



Day 3

- Bit Hydraulics
- Introduction
- Bit nozzle size design
- Optimization of bit hydraulics
- Bit hydraulic horsepower
- Jet impact force
- Exercise
- Hole Cleaning in Vertical Wells
- Introduction
- Particle slip velocity
- Cutting transport ratio
- Factors affecting hole cleaning
- Empirical correlations
- Exercise
- Hole Cleaning in Directional Wells
- Same rules don't apply
- Cutting transport in directional wells
- Difficulties in hole cleaning at different angles
- Factors affecting hole cleaning in directional wells
- Flow rate requirements for hole cleaning in different hole conditions
- Exercises
- Swab and Surge
- Definitions
- Burkhardt method
- Mitchell method
- Recommended speeds for running in hole and pulling out of hole
- Exercises





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